Video Transcript

Video 2: The Evolution of Big Data Architecture and an Introduction to Spark (07:06)

Prior to the web, the generation of data was quite different. You have pretty structured generation that could fit neatly, or at least most of it, into rows and columns. It was a world where you had low velocity, variety, and volume. However, once the web comes along, we really have an explosion of data sources, some of which are still being added today. We certainly had more velocity and more volume, and that didn't fit neatly into that vision of the world that existed prior to it, that world where all of the data went into tables, into rows, and columns and relationships between these tables.

What we had was a big data problem, one where we had velocity or high velocity, variety, and, volume. And companies at the time we're running into this problem, running into this problem of how do we store this tremendous amount of data, this variety and these, this velocity that we have. And so Google in, in the time broke new ground, introduced new innovations with three papers that are widely cited. The very first one is the GFS, the Google file system where they could distribute this oncoming flood of information onto a fabric of commodity servers. So, leveraging low cost hardware, being able to overlay a fabric that could address the diversity, they're the failures that could occur and provide a robust system. On top of that, they added MapReduce, a programming model, and there were other papers that are widely cited such as big tables and others.

Now that was a proprietary system, but the publications detailed well the work that had been done, and so the community was able to leverage that and build an open source version of these ideas. The very first layer was called the Hadoop Distributed File System. The second layer was called the same thing, MapReduce. And on top of that, you had a resource negotiator. Now as you can imagine, this was a great step forward, it broke ground, it changed the capabilities, it provided a tremendous resource that was open source. You didn't have to pay for licenses, you could leverage it. It broke away as well from the cookie cutter solutions of data warehouses. However, as tends to happen, once you take a big step forward, you find you need additional

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things and so there were a number of challenges. One of them was this was open source hardware. It was hard to maintain, it was hard to version, it was hard to evolve. The other thing as well is that although MapReduce was a great programming model, it wasn't necessarily what all industry and certainly verticals were used to, especially within the data community.

Another problem as well was that a speed, reading and writing intermediate results to disk, as is often done in the MapReduce model, was one where you had, depending on the type of loads and work that you were doing, pretty slow performance. This reading to disk versus writing to memory is a tremendous difference when it comes to speed. It's like going down to your gas station or going all the way to the moon or further. And so as you can see here, I'm going to expand a little bit more on why MapReduce was not enough. Ultimately, there were a lot of collisions with the knowledge that already existed in organization, the practices that had already established. And in this case, I'm referring here to the very first bullet, that of SQL, where that had been adopted by the data industry or the data practice as the model when it came to querying data. And so asking them to do MapReduce was really difficult when it came to getting those communities to buy in.

Another is that machine learning wasn't integrated in a way in which it made it convenient to use. And so as you can see there, the Hadoop community adds a number of packages there, Hive, Storm, Impala, and so on, that address many of these other needs. However, the result is that you have even more complexity, even more difficulty in being able to simply install these packages and get a running system. And so there's really a need for a new platform. And that is precisely what Spark focuses on, something that is easier, provides more interfaces, more programming model, it's simpler. It is only one platform now, this multitude of pieces that you have to bring together and it is faster. It writes those intermediate results to memory instead of to disk. More than that, it also decouples storage and compute.

So, you can see here from this illustration, you don't only have to go to Hadoop, as you can see there, the little elephant, you can also go to many additional data sources. And so this turns out to be a very nice architectural abstraction to be able to work with the ecosystem of sources that may be available. And so ultimately they achieve very much that goal. As you can see here, you have a number of different packages for different types of workloads. You have SQL, you have data frames, which is rapidly emerging as one of the default go to structures when it comes to Python in that community, you have streaming data that you can leverage as well. A lot of

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efforts go to that especially as you near or your approach real time, being able to leverage the data real time. You have machine learning algorithms and libraries built into the platform as well and even graph processing.

Now all of that is on top of an engine as you can see there, and it is possible to program against this platform with multitude of languages. There you can see SQL, one of the most important ones, you can see Python certainly in the data community that is key today, as well as some other ones that you may want to use such as R and Java. And so as you can see here, Spark really addresses the need of a new generation of data needs of big data, whereas Hadoop really was a big step forward, certainly a stepping stone, a building block that continues to be used today. However, Apache has really brought that generation forward and created really a very performant, very fast, easy to use new platform that has made a tremendous impact in the big data world.

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